

Introduction

Introduction to NLP — MSc. DH ENC-PSL

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Table of Contents

- 1 Natural Language
- 2 Natural Language Processing
- 3 NLP in this class
- 4 Modern NLP as an Empirical Science
- 5 Tokenization
- 6 Conclusion

1 Natural Language

1.1 What is Natural Language?

1.1.1 Definition

Natural Language

A language that has evolved organically in a community, as a method of communicating between people.

- Modalities: Spoken, Written, Signed
- Examples: English, French, Arabic, Chinese, Samoan, ASL, etc.

1.1.2 In opposition to *Formal Language*

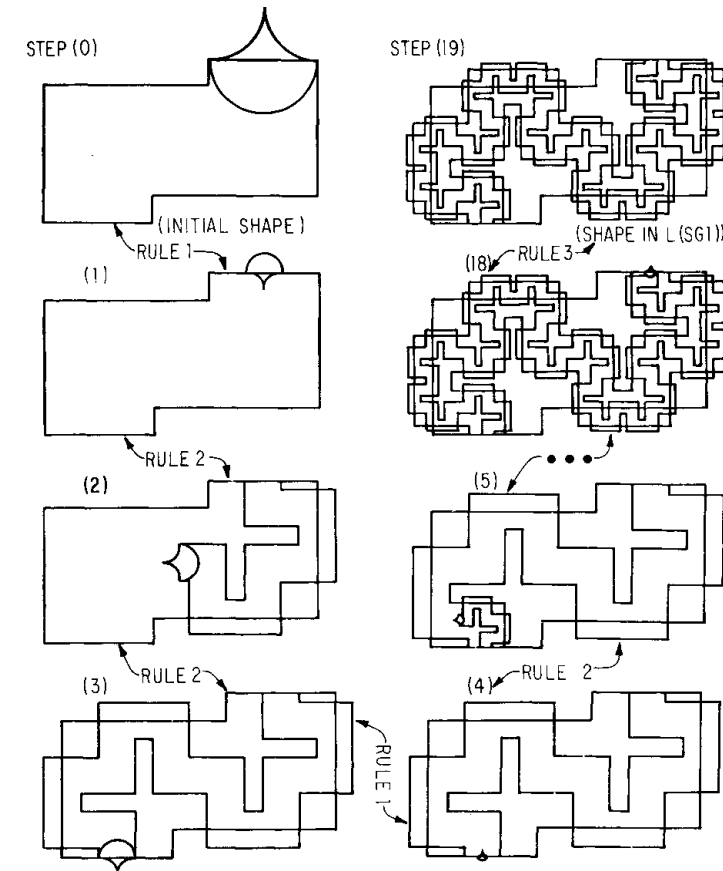
A formal languages is a set of finite symbols (strings) *designed* to avoid ambiguity.

- Examples: programming languages (e.g.; Python), mathematical notations, chemical formulas, etc.



Core NLP challenge

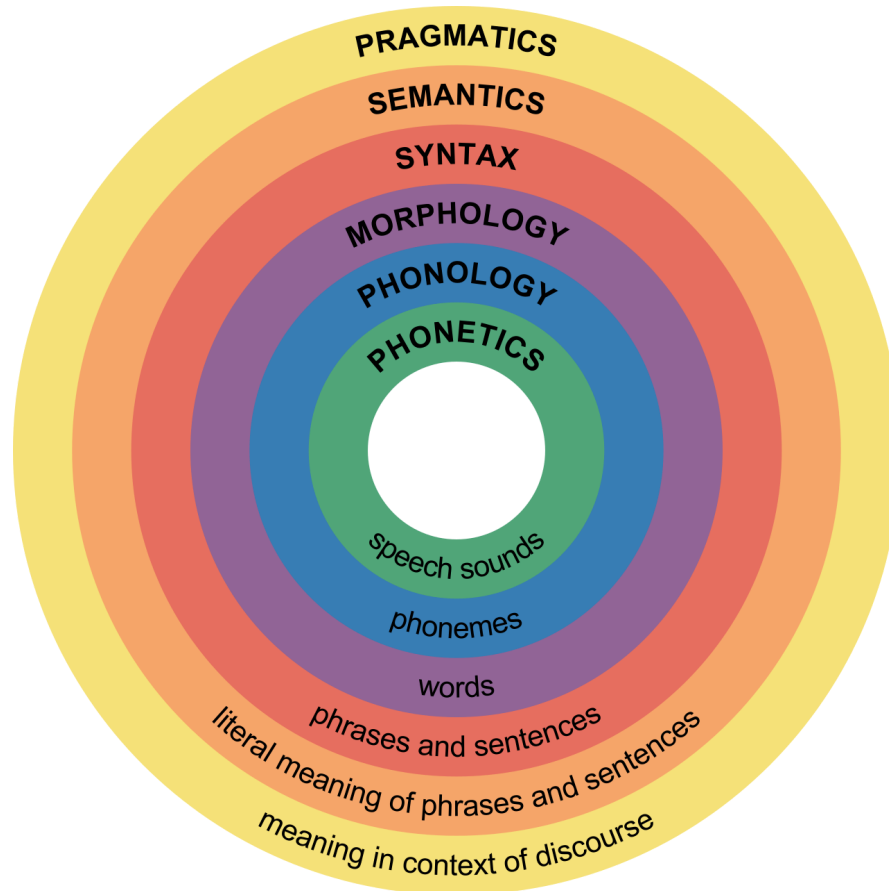
Use deterministic code (formal) to process fluid, non-deterministic natural language.



Generative specification of Stiny and Gips (1971)'s paintings class. Fig 6-1 extracted from (*Stiny and Gips 1971*).

1.2 Linguistics (crash course)

1.2.1 Levels of Linguistic Analysis & Ambiguity



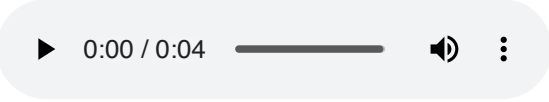
- Linguistic analysis can be decomposed into a hierarchy of several levels.
- Each level reveals different aspects of language.
- Ambiguity exists at every level!

Major levels of linguistic structure. *James J. Thomas and Kristin A. Cook (Ed.) derivative work McSush, via Wikimedia commons.*

1.2.2 Phonology

Focus: System of sounds (*phonemes*) and their representations.

Examples:

- “g'ɪv 'piːz 'eɪ 'tʃans !” vs. “g'ɪv 'piːs 'eɪ 'tʃans !” (IPA transcripts)
- “Un éléphant, marrant”  (see also [Mondegreen](#))

NLP tasks: Speech Recognition, Text-to-Speech (TTS)

1.2.3 Morphology

Focus: Formation and internal structure of words, how smaller units of meaning (*morphemes*) combine.

Examples:

- unhappiness
- → un- + happy + -ness
- uniteable
- un- + [tie + -able] vs. [un- + tie] + -able

NLP tasks: Lemmatization, stemming

1.2.4 Lexical / Part-of-Speech

Focus: study of the meaning and relationships of words

Examples:

- “*Time flies like an arrow; fruit flies like a banana.*” (Groucho Marx)
- “*I made her duck.*”

NLP tasks: Word Sense Disambiguation, Word embedding

1.2.5 Syntax

Focus: Sentence structure, rules for combining words into grammatical phrases.

Examples:

- I saw someone with a telescope.

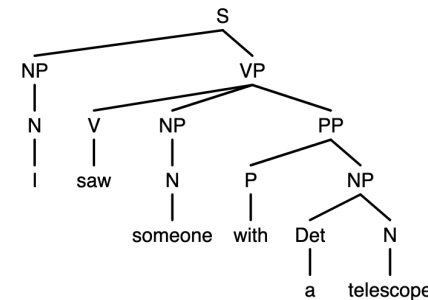
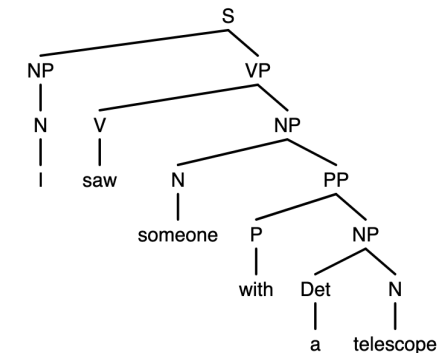


Fig A1.11 and A1.12



from (Catherine Anderson 2022).

NLP tasks: Dependency parsing, POS-tagging

1.2.6 Semantics

Focus: the meaning in language, what is the message being communicated

Examples:

- *“In this country, a woman gives birth every 15 minutes....*
- *... Our job is to find that woman and stop her.”* (Groucho Marx)

NLP tasks: Natural Language Inference (NLI), broadly Natural Language Understanding (NLU)

1.2.7 Pragmatics

Focus: the way sentences with their semantic meanings are used for particular communicative goals

Examples:

- *Can you pass the salt?*

NLP tasks: dialogue systems, hate speech detection

1.3 Empirical behavior of NL

1.3.1 Content words & Function words

Vocabulary can be broadly split into two distinct functional classes:

Class	Examples	Role	Statistical Behavior
Function Words	<i>the, of, and, in, to, is</i>	Grammatical relationships	High frequency, small fixed set (closed class)
Content Words	<i>language, happy, analyze</i>	Primary semantic meaning	Low frequency, unbounded set (open class)

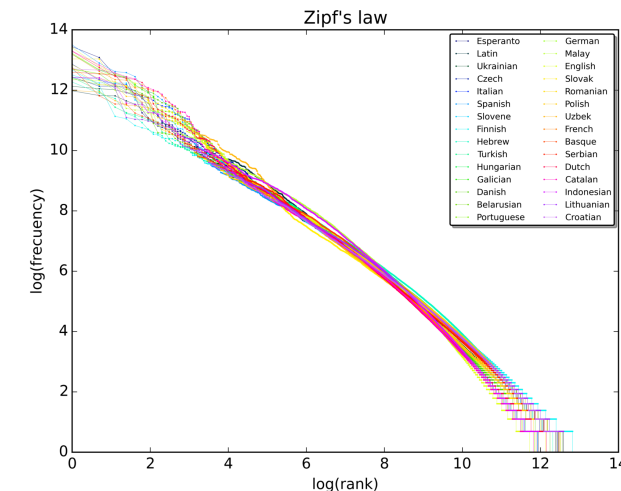
Note: most of the time we are not very interested in function words (stop-words), but they can also reveal interesting aspect (see [Chung and Pennebaker 2011](#)).

1.3.2 Zipf's Law

💡 Zipf Law

The frequency of a word (f) is inversely proportional to its rank (r): $f(r) \propto \frac{1}{r}$.

- **Top 10–20 words** account for up to **50% of all words** in a text.
- **The “Long Tail”**: most unique words occur only once or twice.



Zipf's Law across languages. *SergioJimenez*, via *Wikimedia commons*.

1.4 So, why is NLP hard?

1.4.1 The Core Challenges

1. **Ambiguities** at every linguistic level of analysis
2. **Sparsity (Zipfian Tail)**: most words are rare => most words (combinations) are new
3. **Infinite Expressivity**: infinite way to form sentences with a finite vocabulary
4. **Context Dependency**: meaning is rarely self-contained
5. **Linguistic Variation**: changes across time (diachronic), geography (dialects), social background (sociolects), mediums (tweets vs. news vs. spoken, vs. ...)

2 Natural Language *Processing*

2.1 A quick history

2.1.1 Symbolic Era (~1950s-1990s)

2.1.1.1 Rationalism

Core Hypothesis

Language is governed by explicit, mathematical, and generative grammar rules ([Chomsky 1957](#)).

- **Approach:** Hand-crafting dictionaries, morphological rules, and Context-Free Grammars (CFGs).

→ Deductive reasoning: experts directly encode linguistic knowledge into rule-based computer programs.

2.1.1.2 Important milestones

- Georgetown-IBM Experiment (1954)
 - Machine Translation
 - Translated 60 Russian sentences to English
 - Using 6 grammar rules and 250 vocabulary items.
- ELIZA (Weizenbaum, 1966)
 - Dialogue system
 - Pattern matching on words
 - I need X → What would it mean to you if you got X?
- “five, perhaps three years hence, interlingual meaning conversion by electronic process in important functional areas of several languages may well be an accomplished fact.” (Dr. Dostert, 1954)
- ELIZA effect

2.1.2 Statistical Era (~1990s-2010s)

2.1.2.1 The Empiricist turn

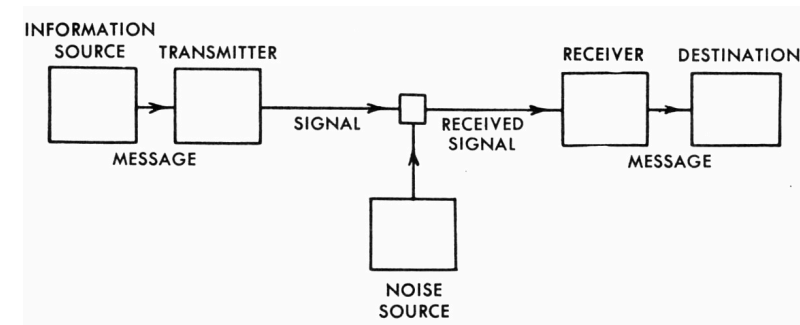
“Every time I fire a linguist, the performance of the speech recognizer goes up.”

— Fred Jelinek (IBM), 1988

💡 Paradigm Shift

Move from hand-crafted expert rules to statistical/probabilistic approach: **counting events in large text corpora.**

Very influenced by *information-theoretic* view of language.



Shannon-Weaver model of communication (Shannon 1948).

2.1.2.2 Classical Machine Learning and Feature-Engineering

- Statistical Machine Translation
- Classification tasks
- Discrete representation
- Rise of evaluation practices and benchmarking

→ Feature-Engineering: practitioners determine discrete rules (features), ML handles the task

2.1.3 Neural Era (~2010s—)

The *AlexNet effect* ([Krizhevsky, Sutskever, and Hinton 2012](#)): sparked Deep Learning (DL) boom!

2.1.3.1 Representation Learning

Key Breakthrough

Learning continuous, dense vector representations (**Embeddings**) end-to-end without manual feature extraction.

- Word2Vec ([Mikolov et al. 2013](#)): words = vectors
- Recurrent Neural Networks (RNNs): language is sequential, let's process it that way (maintain a 'hidden state')

2.1.3.2 The **transformer** revolution & foundation models

“Attention is all you need”! ([Vaswani 2017](#))

- Introduction of “self-attention” mechanism (“look back” at specific parts of inputs at every steps)

→ BERT ([Devlin et al. 2019](#))

→ GPT-series ([Radford et al. 2018](#))

→ etc.

The era of foundation models: Moving from single specialized models per task to massive, general-purpose (tunable) systems.

2.2 NLP in Practice

2.2.1 Common NLP Tasks (source: 🤗)

- Analysing words
 - POS-tagging, Named Entity Recognition, **Word Sense Disambiguation**, ...
- Analysing sentences (to documents)
 - **Sentiment classification**, **Topic Modeling**, Natural Language Inference ...
- Retrieving Information
 - Extracting information from document, **Ranking similar documents**, ...
- Generating text content
 - **Completing Prompt**, **Filling blanks**, ...
- Sequence to sequence generation
 - Machine Translation, Text Summarization, ...

2.2.2 DH Applications

- Analyze style, authorship, ideology across corpora.
 - Track semantic change over time.
 - Support historical text restoration or OCR correction.
 - Build metadata enrichment pipelines.
 - Use models as *tools for exploration*, rather than oracles.
 - ...
- ... but also the other way around: use DH to study LMs.



3 NLP in this class

3.1 Content

3.1.1 Introduction to modern NLP models and methods:

- Language Representation
 - Tokenization, Word representations, Semantic Spaces
- Language Modeling
 - Foundations, n-grams, Transformer, Pre-trained Language Models
- Discovering Structure
 - Retrieval, Topic Modeling
- Inferring Patterns
 - Fine-tuning, Inference, Classification
- Generative LLMs*
 - Prompting, Automating, Evaluating

3.2 Logistics

3.2.1 Calendar

No class on:

- 28th of October
- 11th of November (non working day)

End of the class: 18th of November (included)

3.2.2 Material

Contact: noe.durandard@psl.eu

GitHub Repository:  [d-noe/NLP_DH_F26](https://github.com/d-noe/NLP_DH_F26)

Hands-On: Google Colab, Local Machines, Binder, ...

3.2.3 References

- Always included at the end of the slides
 - some additional (optional) reading
- One book to rule them all: *Speech and Language Processing*, by Jurafsky and Martin (2026)
 - draft version freely accessible online

3.3 Assessment & Grading (T. Poibeau)

3.3.1 Practical & Critical Assignment

Submission: TBD (early- to mid- January 2027)

Length: ~5 pages (**7 max.**) · reasonable font & margins

Start from a **corpus** (preferably your **M2 thesis corpus**) and address a relevant **NLP/TAL question** with a clear objective.

1. **Implement** a solution

Existing code and/or your own approach · **justify your choices** · no off-the-shelf analysis

2. **Evaluate** the results

Qualitative + quantitative where appropriate · measures suited to the **question & data**

3. **Discuss critically**

Results & limitations · **alternatives, improvements & extensions**

3.3.2 Deliverables

Question → Solution → Justification → Evaluation → Critical discussion

Report: ~5 pages · 7 max.

Bibliography: short but compulsory · ≤ 5 references · only sources **actually** read

Presentation: brief final-class presentation + questions on your **approach and results**

Generative AI: explicitly state **how it was used** and follow **school guidelines/charter + disciplinary best practices**

Remember:

A fully finished solution is not absolutely required.

What matters is a methodical, justified & critical approach.

4 Modern NLP as an Empirical Science

4.1 NLP has become data-based

4.1.1 Reminder: The Empirist shift

- **Epistemic Shift:** Language processing becomes an inductive, data-driven science.
- **Hypothesis Testing & Evaluation:**
 - Models are trained (supervised, self-supervised, unsupervised) on corpus data.
 - System performance is measured against empirical benchmarks (*gold standards*).

4.1.2 Core scientific objects

1. **Corpora:** The sampled collection of raw linguistic observations.
2. **Annotations:** The human-interpreted metadata and structural labels mapped onto text.

4.2 Corpus

4.2.1 Definition attempt

 Corpus (pl. corpora) — from ([Pustejovsky and Stubbs 2012](#)).

a collection of **machine-readable texts** that have been produced in a **natural communicative setting**.

They have been **sampled** to be:

- *representative*, the content of the corpus can be generalized to that variety; and
- *balanced*, with respect to particular factors, for example by genre—newspaper articles, literary fiction, spoken speech, blogs and diaries, legal documents.

4.2.2 Text and beyond: multimodal corpora

Not much discussed in this class, but NLP also deals with, e.g.;

Textual & Document Variants

- **Historical Manuscripts:**
OCR/HTR outputs containing visual layouts, noise, and script variations.
- **Social Media Data:** Interspersed with images, emojis, memes, and structural metadata.

Beyond Text

- **Spoken & Acoustic Transcripts:**
Prosody, pauses, speaker alignments, and audio signals.
- **Vision-Language Corpora:**
Images paired with captions or spatial bounding boxes.

4.2.3 Opportunistic vs. purpose-driven data collection

The construction of a NLP research project & corpora selection:

Corpus $\rightarrow$ RQ

1. Come across available data
2. Devise research question

RQ $\rightarrow$ Corpus

1. Start from a research question
 2. Build an appropriate corpus to tackle the RQ
- 2') [Middle-ground:] look for appropriate data (+ expand)

4.2.4 Corpora are samples

- A corpus **cannot** represent the whole linguistic universe: finite sample shaped by availability, selection criteria, collection heuristics, etc.
- **Selection Bias:** These processes favor specific demographics, genres, historical moments, ...

! Important

Data is not a passive, objective reflection of the world. A corpus is a *constructed artifact* subject to sampling biases and human interpretation.

4.3 Annotation

4.3.1 Learning patterns require explicit labels

In **supervised machine learning**, algorithms learn mapping functions between input features ($\approx$ corpus: X) and outputs (target labels: Y):

$$f(X) \rightarrow Y$$

! The fabrication of Truth?

Most of the time, labels are created through human interpretation. “Gold Standard” or “Ground Truth” is a **socially constructed consensus**, rarely an objective physical property of the text.

4.3.2 Structural levels of annotations

- Word and subword level
 - (Morphological parsing)
 - (POS-tagging)
- Span level
 - Continuous (Named Entity Recognition (NER))
 - Discontinuous (Syntactic units, idiom components)
- Relational level
 - Directed relations (Relation extraction, Knowledge graphs)
 - Equivalence chains (Coreference resolution)
- Document level
 - (Sentiment analysis, Genre classification)

Le pauvre garçon voulut paraître fort, et il répéta plusieurs fois :

[– Oui..., du courage !]_{DS}

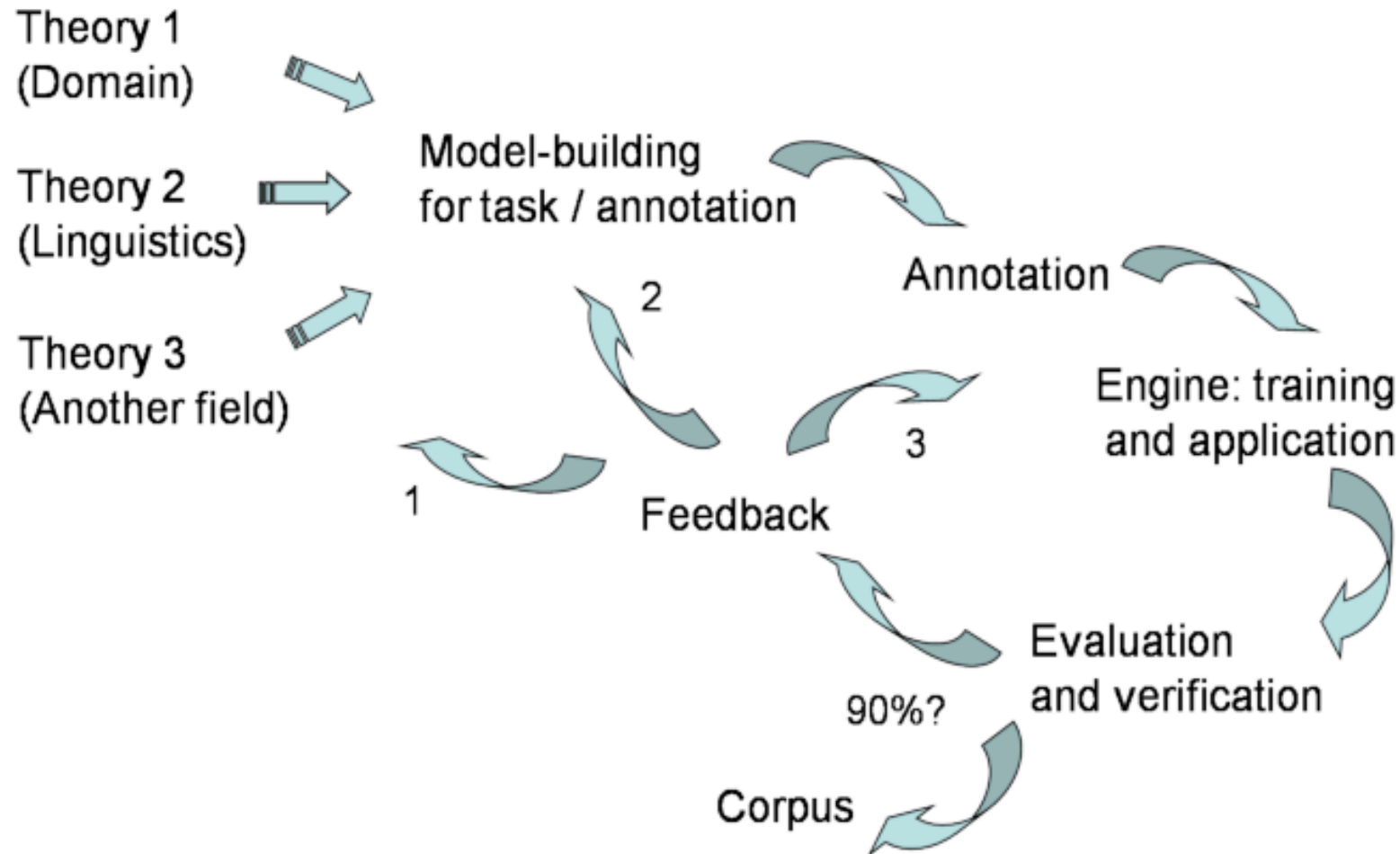
[– Eh bien,]_{DS} s'écria le bon-homme, **[j'en aurai, nom d'un tonnerre de Dieu ! Je m'en vas la conduire jusqu'au bout.]**_{DS}

The poor fellow tried to show himself brave, and repeated several times. **["Yes! courage!"]**_{DS}

["Oh,"]_{DS} cried the old man, **["so I will have, by God! I'll go along o' her to the end!"]**_{DS}

Excerpt of *Madame de Bovary* (Flaubert, 1856).
Extracted from ([Durandard et al. 2023](#)).

4.3.3 The annotation pipeline



The generic annotation pipeline, extracted from ([Hovy and Lavid 2010](#)).

4.3.4 Interannotator agreement

Cohen's Kappa (κ) measures agreement between 2 *annotators* on categorical tasks, correcting for chance agreement:

$$\kappa = \frac{p_o - p_e}{1 - p_e}$$

- p_o : observed agreement
- p_e : expected probability of agreement (how often would annotators agree assuming *independent* annotations)

4.3.5 Advance IAA & rule-of-thumb

Value of Kappa	Level of Agreement	% of Data that are Reliable
0–.20	None	0–4%
.21–.39	Minimal	4–15%
.40–.59	Weak	15–35%
.60–.79	Moderate	35–63%
.80–.90	Strong	64–81%
Above .90	Almost Perfect	82–100%

Rule of thumb to interpret IAA metrics. *Extracted from* ([McHugh 2012](#)).

- **Fleiss' Kappa:**
Generalization of Cohen's κ for > 2 annotators.
- **Krippendorff's Alpha (α):**
Highly flexible metric handling multiple annotators, missing data, and different scale types (nominal, ordinal, interval).

4.4 **Workshop**: Sentiment annotation

4.4.1 Annotation Manual

Please read the instructions for the sentiment labelling task.

You will see different sentences in the notebook. Your task is to decide whether each sentence has a *positive*, *negative*, *neutral* or *mixed* sentiment. For example, “*I like warm summer evenings*” has a *positive* sentiment.

“*Mixed*” label refers to a sentence that has a mixed sentiment. For example, the sentence “*I liked the food, but the waiter was rude*” has a *mixed* sentiment.

Try to be objective with your judgement.

4.4.2 Analysis

The data for the workshop actually comes from the study “Disambiguating sentiment annotation” ([Äyräväinen, Hinds, and Davidson 2025](#)).

Further Reading on DH and Data annotation

- ([Stahel et al. 2025](#)) on perspectivist annotation pipelines.
- ([Krusic 2024](#)) on constructing datasets with soft-label distributions.
- ([Schmidt et al. 2019](#)) on disagreement in literary text analysis.

4.4.3 Alternatives? *Data perspectivism*

Myth One: *One Truth*

Most data collection efforts assume that there is one correct interpretation for every input example.

Myth Two: *Disagreement Is Bad*

To increase the quality of annotation data, disagreement among the annotators should be avoided or reduced.

Myth Three: *Detailed Guidelines Help*

When specific cases continuously cause disagreement, more instructions are added to limit interpretations.

Myth Four: *One Is Enough*

Most annotated examples are evaluated by one person.

Myth Five: *Experts Are Better*

Human annotators with domain knowledge provide better annotated data.

Myth Six: *All Examples Are Created Equal*

The mathematics of using ground truth treats every example the same; either you match the correct result or not.

Myth Seven: *Once Done, Forever Valid*

Once human annotated data is collected for a task, it is used over and over with no update. New annotated data is not aligned with previous data.

The “Seven Myths” of data annotation, by Aroyo and Welty (2015).

Recent developments challenge the traditional label-construction model.

Data perspectivism ([Cabitza, Campagner, and Basile 2023](#)):

- *soft labeling*: probabilistic labels ($\neq$ hard, categorical labels)
- **disagreement = valuable data** ($\neq$ error or noise)

See also [perspectivist data manifesto](#), or workshops such as [NLPerspectives](#).



5 Tokenization

5.1 Motivation

5.1.1 The symbolic gap

- Computers do **not** process words, concepts, or historical nuances directly—they compute on **numbers, matrices, and vectors**.
- **Raw Text in Memory:** A continuous stream of Unicode bytes or ASCII characters.

5.1.2 Definition

Tokenization

the task of separating out or tokenizing words and word parts from running text.
([Jurafsky and Martin 2026](#))

Process is:

Raw text → **tokenize** → Tokens decomposition →
vocabulary lookup → Vocabulary IDs

5.1.3 Is it token=word?

→ What is a *word*?

- cat
- I'm
 - one orthographically, two grammatically
- C U
- San Francisco
- what about languages with no separators?

Ill-defined, ambiguous and context dependent.

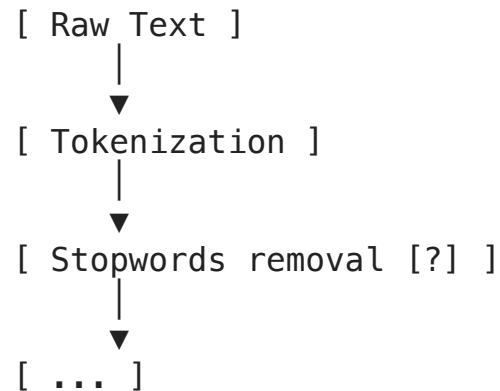
Not really, we'll come back to that shortly (but for simplicity, you can think of token=words most of the time).

5.2 The universal first step

5.2.1 Always tokenize

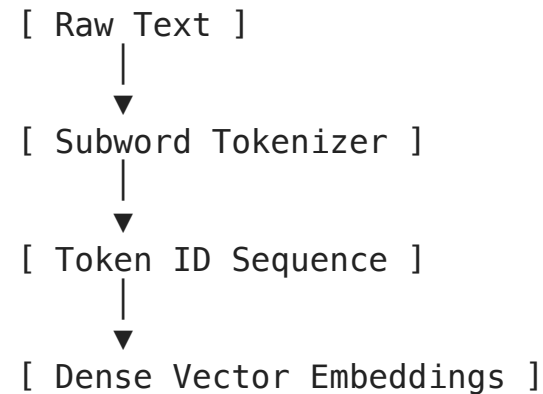
Regardless of historical era or model paradigm, **tokenization is the gateway to every NLP pipeline**:

Traditional NLP Pipeline



- Discrete rule-based or statistical features.

Modern Neural Pipeline



- Continuous vector spaces & Transformers.

5.2.2 Some useful libraries

Different libraries specialize in different tokenization paradigms:

Library	Paradigm	Typical Level	Primary Use Case
NLTK	Rule-based & regex heuristics	Words / Sentences	classical NLP prototyping
spaCy	Dependency-aware tokenization	Words / Morphemes	Production linguistic pipelines (POS, NER)
HuggingFace Tokenizers	Subword statistical models	Subwords (BPE, WordPiece)	Fine-tuning Transformer models (BERT, RoBERTa)
tiktoken	Byte-Pair Encoding (BPE)	Subwords / Bytes	OpenAI LLM models (GPT-3.5, GPT-4o)

5.3 ‘Word’-Tokenizers

5.3.1 Regex Pattern matching

Using regular expressions to isolate 'words' explicitly:

```
1 import re
2 text = "The cat is on the mat, the dog isn't."
3 tokens = re.findall(r"\w+|[\^\\w\\s]", text)
4 # Output: ['The', 'cat', 'is', 'on', 'the', 'mat', ',', 'the', 'dog', 'isn'
```

Limitations: **Out-Of-Vocabulary** (OOV) issues, rely on **whitespace delimiters** (what about other languages: Chinese, Japanese, Ancient Greek *scriptio continua*, etc.?)



Tip

Note that such approaches are still the foundation pre-tokenization step of modern tokenizers!

5.4 Subword tokenization

5.4.1 Byte-Pair Encoding (BPE) (Gage 1994)

Directly learn from the data!

```
1 function BYTE-PAIR ENCODINGS(C:string, vocab_size:int):  
2     V = [all unique bytes in C] # initial set of tokens is characters (bytes)  
3  
4     learned_merges = [] # list of learned merges  
5     while len(V) < vocab_size: # expand vocab (merges) until reaching vocab_s  
6         (a, b) = get_most_frequent_pair(C) # most frequent adjacent pair  
7         new_token = a + b # make new token by concatenation  
8         V += [new_token]  
9         learned_merges += [(a,b), new_token]  
10        update(C, learned_merges) # update corpus: replace each occurrence of (a  
11  
12    return V
```

5.5 Hands-on Exercise

6 Conclusion

References

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